Daizhi Liao

Undergraduate researcher · Reinforcement Learning & Embodied AI · PhD applicant Fall 2027

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EDUCATION

Guangzhou University — B.Eng., Network Engineering Expected 2027

Third-year undergraduate. Focus: reinforcement learning, embodied AI, and the empirical auditing of deployed AI systems.

RESEARCH & PROJECTS

TransitTruth — Behavioral-Fingerprint Audit of AI API Relays 2026

Solo design & implementation. Open-source tool that verifies whether an AI API relay actually serves the advertised model, by probing statistical behavior (random-number, token, latency distributions) against baselines with chi-square / KS tests and Bayesian updating. Built a zero-install, fully client-side web demo and a technical report validating same-family discrimination: gpt-4o vs gpt-4o-mini, TVD up to 0.90, 91.3% in-distribution resubstitution posterior (not held-out).
Code & demo: github.com/dafahaha/transit-truth · live demo

embodied-offpolicy-study — Pre-registered Three-Arm SAC Study on Hopper-v4 2026

Soft Actor-Critic implemented from scratch in PyTorch (no Stable-Baselines3), run as a pre-registered three-arm controlled study on Hopper-v4 (CPU, 120k env steps, 3 seeds/arm): Baseline (reward scale 1.0) vs RewardScaled (0.1) vs FixedAlpha (0.1, automatic temperature off, α fixed at 0.2). Mean return ± SD (final window 105k–120k, n=3/arm): Baseline 1085±698, RewardScaled 1118±623, FixedAlpha 1404±632. Welch p: Baseline–FixedAlpha 0.59, Baseline–RewardScaled 0.95, FixedAlpha–RewardScaled 0.61; one-way ANOVA F=0.22, p=0.81 — no statistically distinguishable effect detected (n=3; the t-intervals overlap heavily). At 60k the reward-scaled arm looked ~3× better, but extending to the pre-registered 120k budget and adding the fixed-temperature control overturned that apparent gain as early seed noise: an honest pre-registered null. Raw CSVs committed, curves reproduced by script.
Code: github.com/dafahaha/embodied-offpolicy-study

rl-deploy-bench — Cross-Platform RL Deployment & Benchmarking v1.1.0

Config-driven CLI: export Stable-Baselines3 policies to ONNX/TorchScript, build TensorRT engines with FP16/INT8 quantization, and benchmark latency, throughput, and accuracy across x86 GPU and CPU, with auto-generated HTML reports. Jetson/edge is a target deployment path, not yet measured on device. A reproducible harness for the accuracy–latency tradeoff of quantized RL policies.

PUBLICATIONS & PREPRINTS

- Behavioral Fingerprinting of LLM API Relays — detecting model substitution on AI API relays (double-blind workshop paper). *In preparation.*
- TransitTruth: Extended Technical Report — behavioral probing, chi-square/KS & Bayesian self-consistency, threats to validity (2026). PDF.

OPEN SOURCE — MERGED CONTRIBUTIONS

- **microsoft/TextWorld #377** (merged) — Fixed `env.step` crash on multi-command input; non-greedy score parsing plus regression test.
- **shmuma/ptan #57** (merged) — Fixed Gymnasium API compatibility in experience sources.
- **Algorineko/AgenticArXiv-RL #72** (merged) — Added MIT License, CONTRIBUTING guide, Code of Conduct, and CI.
- **learnsyslab/gym-pybullet-drones #319** (merged) — Replaced `print()` error statements with module logging; converted `exit()` calls to raised exceptions.

Beyond merged work, additional pull requests are **currently under review (not yet merged)** upstream in e.g. vLLM, pandas, google/brax, Farama-Foundation/PettingZoo, and Hugging Face datasets & diffusers.

TECHNICAL SKILLS

RL & learning: PyTorch, Stable-Baselines3, Gymnasium, MuJoCo | **Deployment:** ONNX, ONNX Runtime, TensorRT, TorchScript, INT8/FP16 quantization | **Systems:** Python, C++, Linux, Docker, Git, CI/CD, CUDA